

## Turtle Games technical report

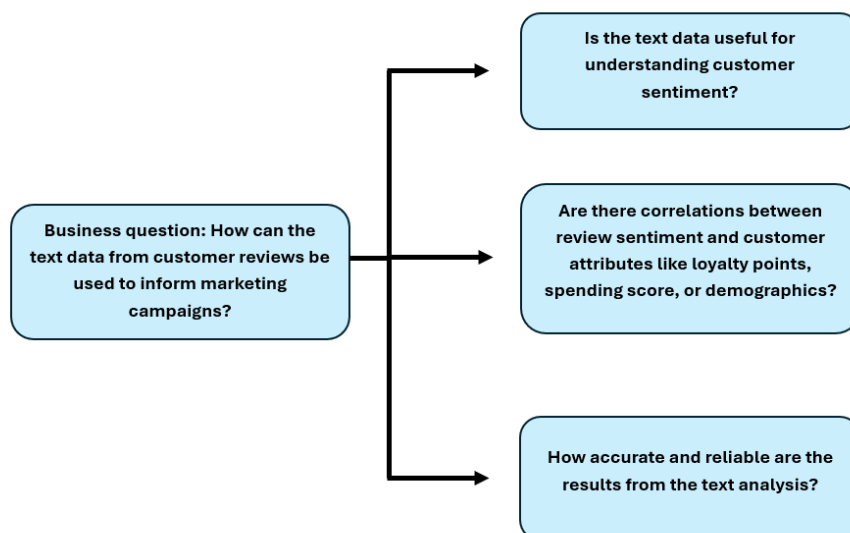
### Background of the business scenario

Turtle Games want to better understand customer behaviour to improve sales performance and identify customer trends. They need to understand:

1. How customers engage with and accumulate loyalty points and whether loyalty points are an important metric.
2. How customers can be segmented into groups, and which groups can be targeted by their marketing department.
3. How text data from customer reviews can be used to inform their marketing campaigns and make improvements to their business.
4. Whether descriptive statistics can provide insights into the suitability of the loyalty points data for predictive modelling.

### Analytical approach

I used an approach like an issue tree structured thinking framework to break the main business questions down into sub questions so that I could complete deeper analysis.



*Fig 1: Example of the structured thinking framework I used with each business question.*

I used functions to improve code readability and reduce repetition which helped save time during my analysis. The Seaborn "colorblind" palette was used as the default palette to enhance accessibility for colourblind people, and the default order was set to descending to clearly highlight the highest and lowest values.

```

# Function to create a count plot.
def count_plot(df, column, title="Count Plot", xlabel=None, ylabel="Count", order=None, palette="colorblind", legend_title=None, **kwargs):

    # Sort the count of the categories in descending order.
    if order is None:
        order = df[column].value_counts().index

    # Create the count plot
    plt.figure(figsize=(10, 6))
    sns.countplot(data=df, x=column, palette=palette, order=order, **kwargs)

    # Get the axes so that the Legend part of this function works.
    ax = plt.gca()

    # Set Labels and title
    plt.title(title, fontsize=14)
    plt.xlabel(xlabel if xlabel else column.title(), fontsize=12)
    plt.ylabel(ylabel, fontsize=12)
    plt.xticks(rotation=45)

    # Add a Legend if there are handles.
    handles, labels = ax.get_legend_handles_labels()
    if handles:
        plt.legend(handles, labels, title=legend_title, loc="upper left", bbox_to_anchor=(1, 1))

    # Save the figure using the title of the plot and show the plot
    plt.tight_layout()

    clean_title = re.sub(r'^\w\s-', '', title.replace("\n", " ").strip().replace(' ', '_').lower())
    filename = f"{clean_title}.png"

    plt.savefig(filename, dpi=500, bbox_inches="tight")

    plt.show()

```

*Fig 2: Example of a function used.*

The dataset was cleaned using the following steps:

- Validating the data using the “validate\_date” function which included a check of the data types and identifying duplicated rows.
- Check for missing (null) values.
- Unnecessary columns were dropped. The “language” and “platform” columns were removed as they each contained only one unique value and therefore did not contribute anything meaningful to the analysis.
- Renaming columns for clarity.
- Check for unique values in the text columns and if needed, the values were renamed for clarity.
- The loyalty points variable was found to be heavily right-skewed. Outlier analysis revealed that 12.92% of customers were outliers. However, these were not removed as the distribution seemed reasonable, likely representing a small group of highly loyal customers. The histogram showed no gaps between bins, suggesting that these high values were not anomalies.

The following are some screenshots from the data cleaning process. Additional screenshots can be found in the Appendix.

```
# Validate the data function.
def validate_data(df):
    # Check the data types.
    data_types = df.dtypes
    print("Data types:")
    print(data_types)

    # Check the data shape.
    data_shape = df.shape
    print("Data shape:")
    print(data_shape)

    # Check for unique values
    unique_counts = df.nunique()
    print("Unique values per column:")
    print(unique_counts)

    # Check for duplicates
    duplicate_count = df.duplicated().sum()
    print("\nNumber of duplicate rows:")
    print(duplicate_count)

    # Summary statistics
    summary_stats = df.describe()
    print("\nSummary statistics:")
    print(summary_stats)

    return data_types, data_shape, unique_counts, duplicate_count, summary_stats
```

Fig 3: validate\_data function

```
# Use the validate_data function to get some summary information about the DataFrame.
validate_data(reviews)

Data types:
gender          object
age             int64
remuneration (k€) float64
spending_score (1-100) int64
loyalty_points  int64
education       object
language        object
platform        object
product         int64
review         object
summary         object
dtype: object
Data shape:
(2000, 11)
Unique values per column:
gender          2
age             45
remuneration (k€) 64
spending_score (1-100) 84
loyalty_points  627
education       5
language        1
platform        1
product        200
review         1980
summary        1432
dtype: int64

Number of duplicate rows:
0
```

Fig 4: Using the validate\_data function and an extract from the output.

```
# Rename the column headers.
reviews.rename(columns={"remuneration (k€)": "income_k", "spending_score (1-100)": "spending_score"}, inplace=True)

# View column names.
reviews.columns

Index(['gender', 'age', 'income_k', 'spending_score', 'loyalty_points',
      'education', 'product', 'review', 'summary'],
      dtype='object')
```

Fig 5: Renaming the column headers for clarity.

When inspecting the values in the rows, I found that excluding the product codes, reviews, and summaries revealed a significant number of duplicated rows. Since it is unlikely that multiple customers would share identical demographics, loyalty points, and spending scores, I concluded that there were individual customers who reviewed multiple products. I therefore created a customer\_id column so I could analyse the data for each individual customer.

|      | A      | B   | C                 | D                      | E              | F         | G        | H        | I       | J                                                                                                     |
|------|--------|-----|-------------------|------------------------|----------------|-----------|----------|----------|---------|-------------------------------------------------------------------------------------------------------|
| 1    | gender | age | remuneration (k€) | spending_score (1-100) | loyalty_points | education | language | platform | product | review                                                                                                |
| 109  | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 3165    | This create your own robot sticker book is exactly what I was hoping it would be! I would definitely  |
| 309  | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 9875    | They do not bond well together. the head of the puppies do not bond well to the body.                 |
| 509  | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 9875    | Can be kind of difficult to play. Rules aren't very clear. Described as being like UNO. UNO is easier |
| 709  | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 2285    | Spread everywhere in my house!                                                                        |
| 909  | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 2326    | Great Game, but cheaply made. Still worth while. If you can get a vintage set go for it.              |
| 1109 | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 5430    | I got this game from a local shop and I must say I wish I would've known this was on amazon becau     |
| 1309 | Male   | 53  | 51.66             | 46                     | 1756           | PhD       | EN       | Web      | 8275    | Great expansion to Lords of Waterdeep. This expansion adds three different levels of play so you      |

Fig 6: Example of an individual customer who wrote multiple reviews.

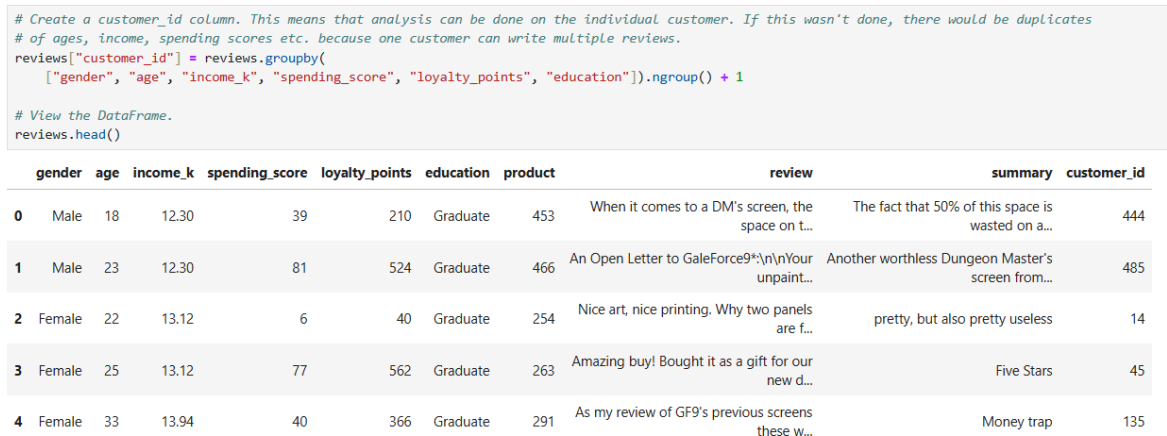


Fig 7: Creating a customer\_id column.

Before modelling, I conducted exploratory data analysis to better understand the data. This helped to identify the variable distributions and potential anomalies that could potentially impact modelling. The following are some screenshots from this analysis. Additional screenshots can be found in the Appendix.

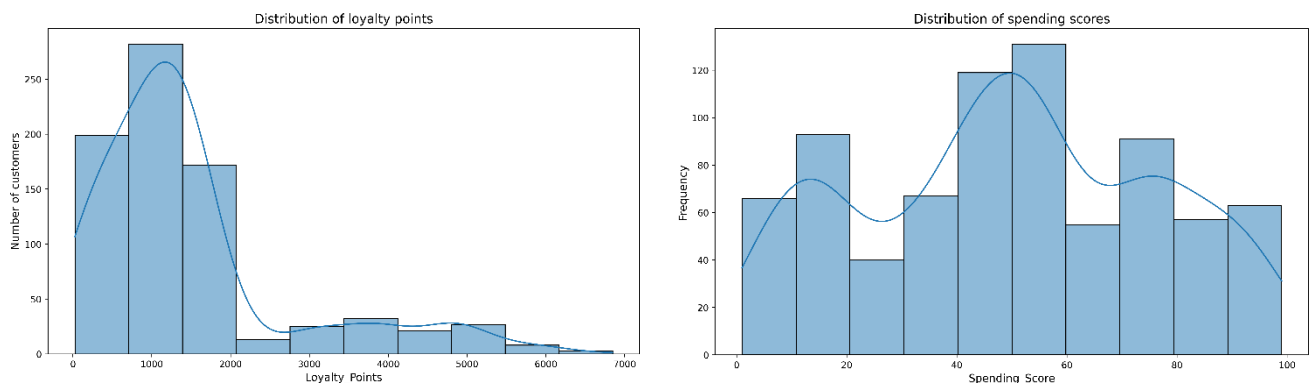


Fig 8: Examples of exploratory analysis of the distribution of loyalty points and spending scores.

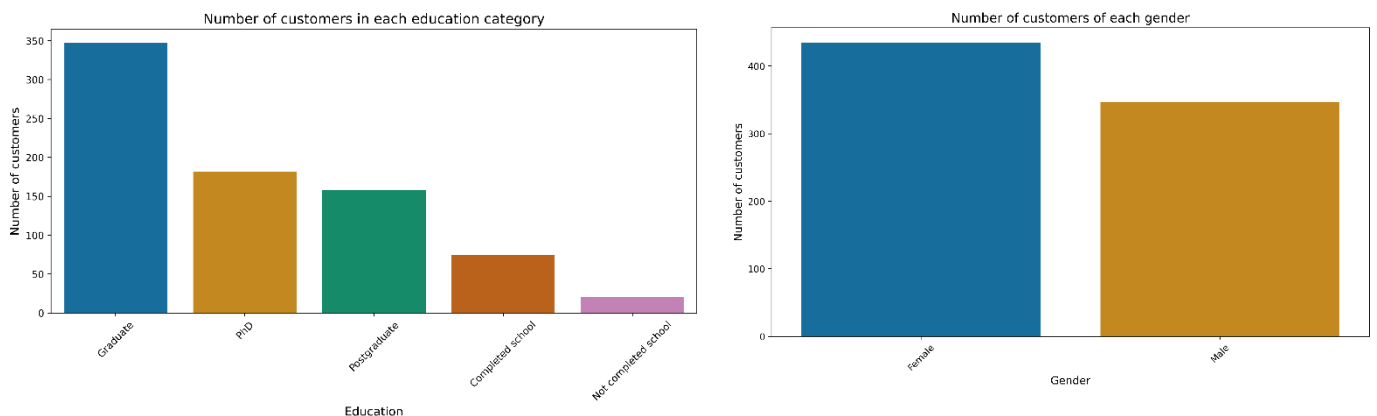


Fig 9: Examples of exploratory analysis of the distribution of customers across education levels and gender.

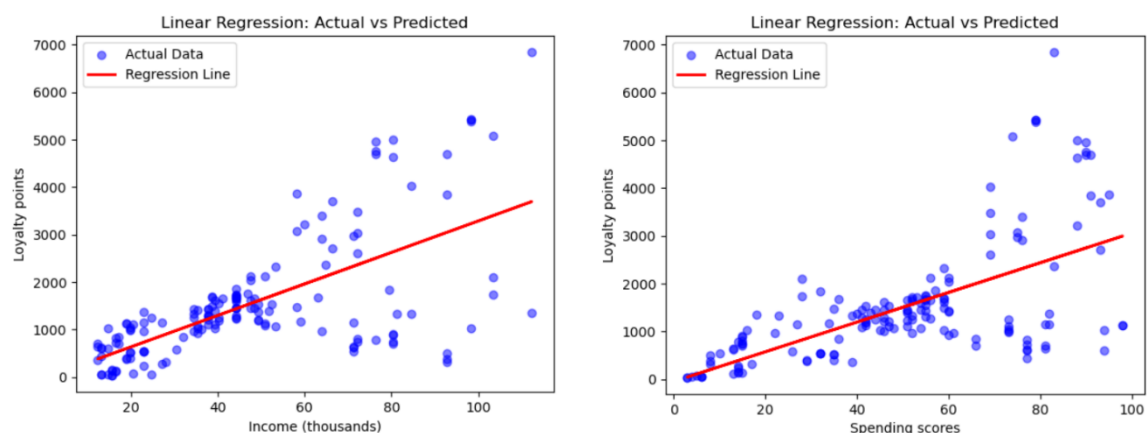
I used simple linear regression first, as a simpler model allows for easier interpretation, avoids unnecessary complexity, and can be more computationally efficient. Sometimes the skew and kurtosis indicated that there was a non-normal distribution or a higher amount than expected of extreme values. As simple linear regression had limited effectiveness, I applied a log transformation to loyalty points and then tried support vector regression. However, due to the lack of a consistent linear relationship, these models performed poorly. In contrast, decision tree regression and classification methods produced lower errors and were a better fit for the data, although there were some issues with potential overfitting.

## Insights

Key insights:

1. High loyalty only occurs when income or spending scores are high; however, a high income or high spending score alone does not guarantee high loyalty.
2. Spending scores and income are the strongest predictors of loyalty points, with age having a minor influence.
3. Linear models were overall a poor fit for the data, capturing some patterns but failing to generalise across all of the data.
4. Customers can be segmented into five distinct groups based on their spending score and income.
5. Review sentiment is mostly positive, with 90% of reviews classified as positive and 78% as very positive.

The following are screenshots that support some of the insights stated above. Additional screenshots can be found in the Appendix.



*Fig 10: Scatterplots with the regression line showing how the model is a better fit for lower income and spending scores, but is a poor fit for higher income and spending scores. This distribution of the data led to large RMSE's and the violation of assumptions like heteroscedasticity and non-normality of residuals.*

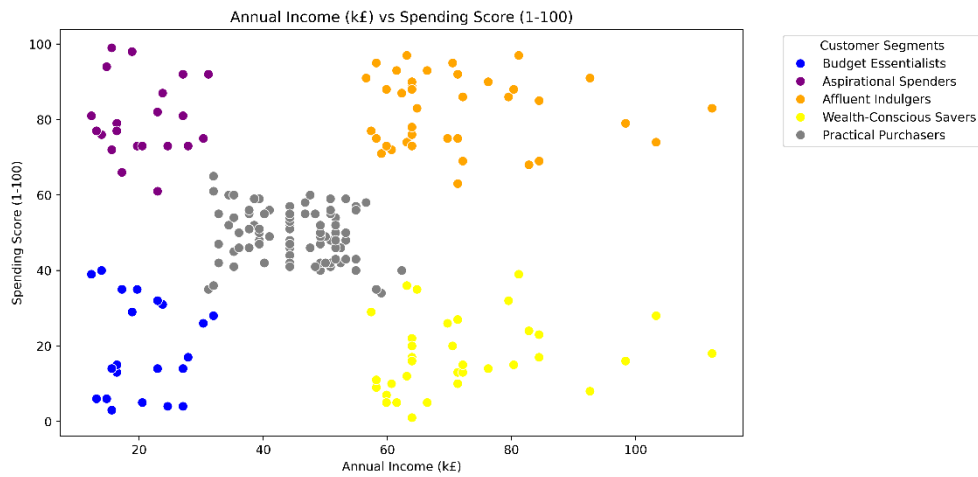


Fig 11: *k*-means clustering effectively segmented customers into five distinct segments using spending scores and customer income.

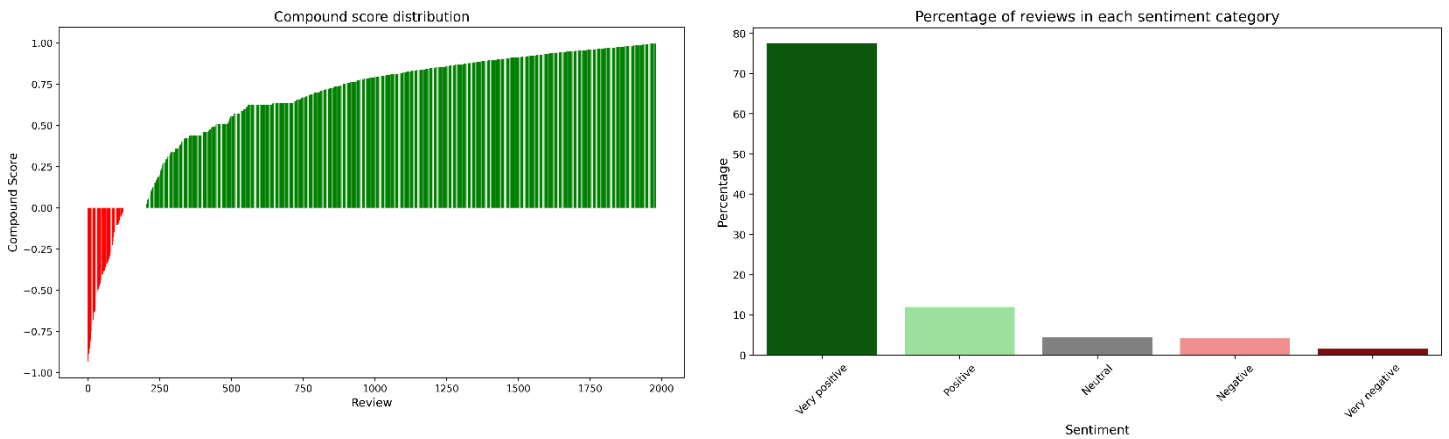


Fig 12: Bar plots showing that most review sentiments are positive, and the majority are very positive.

## Recommendations to the company

Recommendations include:

1. Target high income customers in marketing campaigns, as they are more likely to be highly loyal – although not all of them are. Particularly target those who have high income and spending (“Affluent Indulgers”) as they will bring in lots of revenue for the company. Since some in this group have low loyalty points and spending scores, consider using surveys to better understand their product preferences. Focus on advertising high quality products that align with their interests to encourage greater engagement and loyalty.
2. Use the five customer segments to create targeted marketing strategies aimed at encouraging continued purchases and increasing loyalty. For example, for “Budget Essentialists” (low spending and income), promote low cost items, special offers, and allow them to redeem loyalty points on products.

3. Although most review sentiments are positive, negative reviews can negatively impact sales. Prioritise gathering feedback from high income customers – who are more likely to be highly loyal customers – through surveys or follow-up calls to understand their concerns. Provide meaningful incentives, such as discounts or complimentary gifts, along with flexible return options to rebuild trust and encourage repeat purchases.

### **Limitations and recommendation for further analysis**

The analysis was constrained by missing data, limiting the strength of the conclusions and recommendations. Additional data would improve the robustness and depth of the analysis.

Recommended data for future analysis includes:

- Data on the customers who didn't write reviews so that the company can better understand all of its customers.
- Details on how customers accumulate loyalty points and what they can do with these loyalty points. If they can spend these loyalty points then loyalty points might not necessarily correlate with the amount of loyalty to the company.
- Data on how long these customers have been customers of the company, how many products they've purchased, and when they bought these products could allow for a more accurate picture of customer loyalty. For example, a long-term customer with sporadic purchases may have higher loyalty points but lower actual loyalty to the company than a new, frequent buyer.
- Data on the product categories the customers bought from so analysis can be done on which products the most and least loyal customers tend to buy.
- Income and loyalty points, loyalty points and spending scores, and spending scores and loyalty points all had the same correlation (0.63). This could indicate that there is another variable that isn't in this data set that is influencing these variables e.g. amount spent, the number of products bought, or how often customers purchase items.

## Appendix

### Data cleaning and wrangling

```
# Check for missing values.
reviews.isna().sum()
```

```
gender          0
age             0
remuneration (k£) 0
spending_score (1-100) 0
loyalty_points  0
education       0
language        0
platform        0
product         0
review          0
summary         0
dtype: int64
```

Fig 1: Check for missing values (nulls).

```
# Drop unnecessary columns.
reviews.drop(["language", "platform"], axis=1, inplace=True)
```

```
# View column names.
reviews.columns
```

```
Index(['gender', 'age', 'remuneration (k£)', 'spending_score (1-100)',
      'loyalty_points', 'education', 'product', 'review', 'summary'],
      dtype='object')
```

Fig 2: Dropping unnecessary columns.

```
# Check unique "education" values
reviews["education"].unique()

array(['graduate', 'PhD', 'diploma', 'postgraduate', 'Basic'],
      dtype=object)
```

Fig 4: Checking the unique values in the text columns.

```
# Rename values in the "education" column.
reviews["education"].replace({"Basic": "Not completed school", "diploma": "Completed school",
                             "graduate": "Graduate", "postgraduate": "Postgraduate"}, inplace=True)

# Check unique "education" values
reviews["education"].unique()

array(['Graduate', 'PhD', 'Completed school', 'Postgraduate',
      'Not completed school'], dtype=object)
```

Fig 5: Renaming the values in the "education" column.

```
67 # 1. Remove columns that aren't needed for this analysis:
68 # "language", "platform", "review", "summary".
69
70 # Drop columns
71 reviews_cleaned <- select(reviews, -language, -platform, -review, -summary)
72
```

Fig 6: An example of data cleaning in RStudio – Removing the columns not needed for the analysis.

```
77 # 2. Rename columns: remuneration..k.. and spending_score (1-100).
78
79 # Rename columns.
80 reviews_cleaned <- reviews_cleaned %>%
81   rename(income_k = remuneration..k.,
82          spending_score = spending_score..1.100.)
83
```

Fig 7: An example of data cleaning in RStudio – Renaming the columns.



## Exploratory analysis

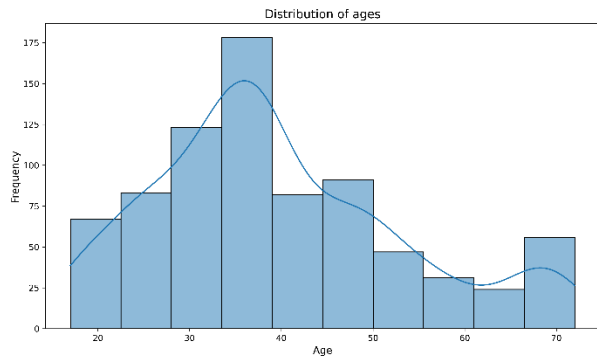


Fig 8: The distribution of ages.

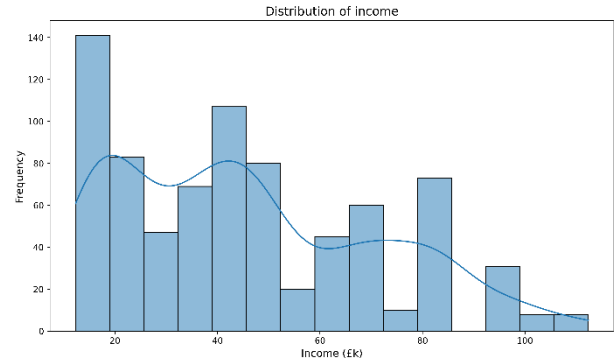


Fig 9: The distribution of income.

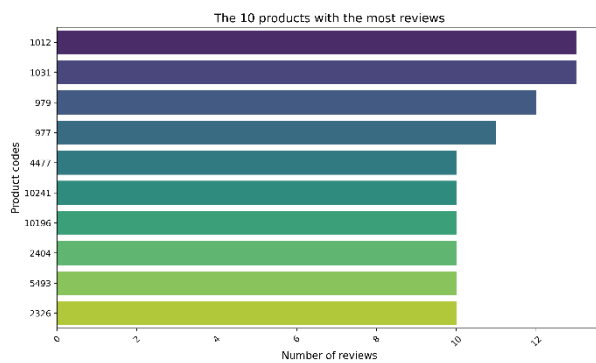


Fig 10: The 10 products with the most reviews.

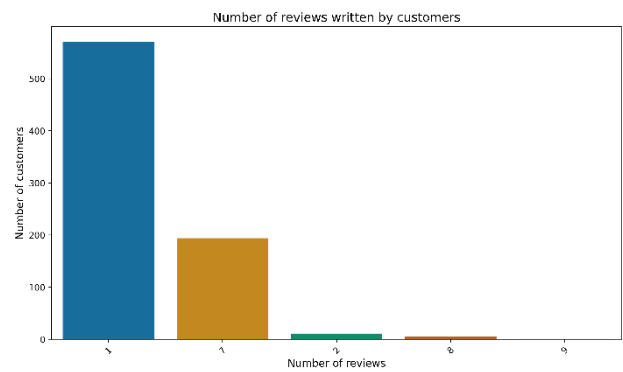


Fig 11: The number of reviews written by customers.

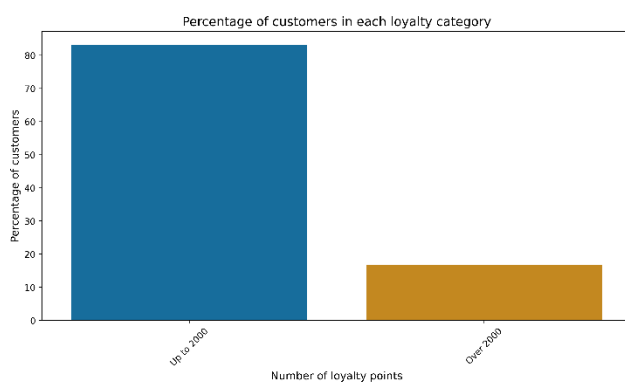


Fig 12: The percentage of customers with less than 2000 loyalty points and more than 2000 loyalty points.

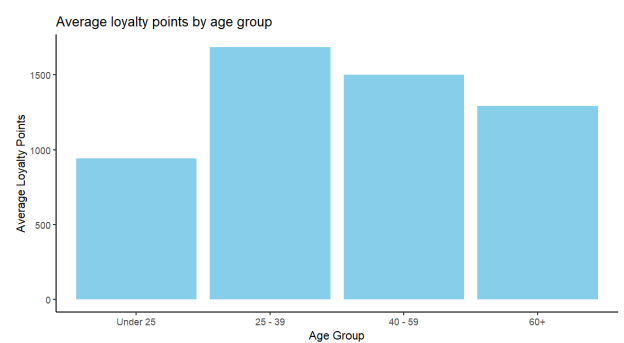


Fig 13: The average number of loyalty points by age group (RStudio).

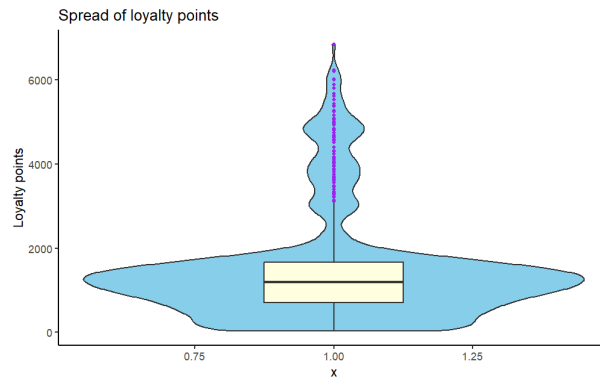


Fig 14: A violin plot (with a box plot) showing the distribution of loyalty points (RStudio).

## Insights

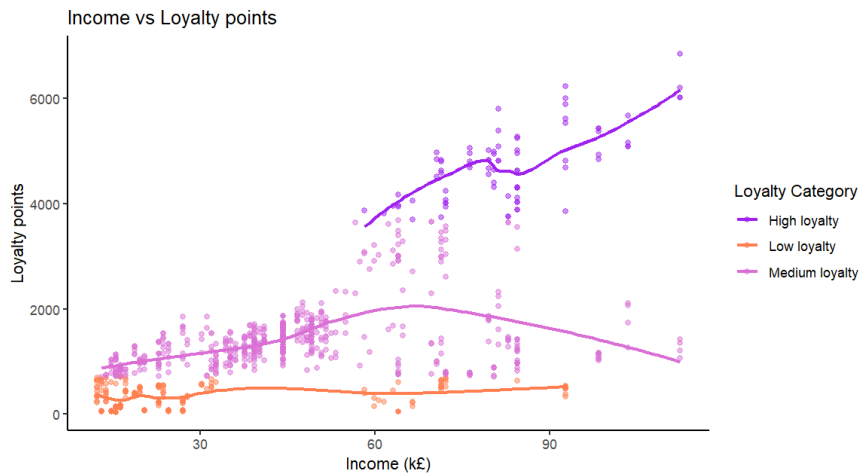


Fig 15: Scatterplot with splines showing income vs loyalty points grouped by loyalty category. High loyalty only happens when income is over about £60k.

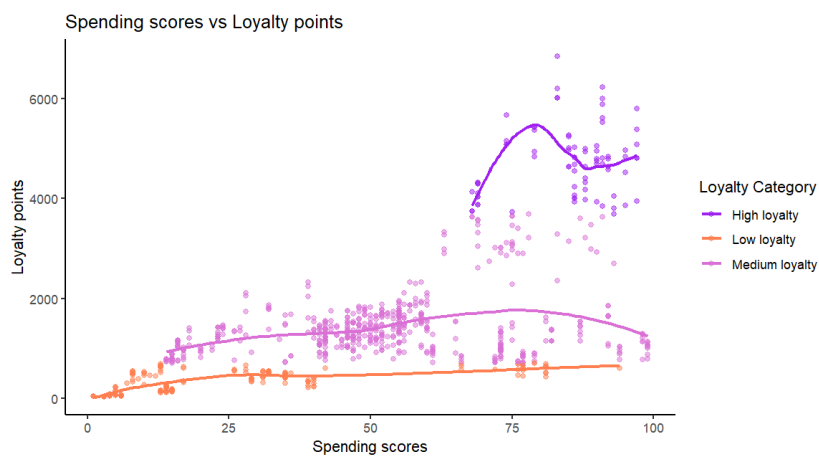


Fig 16: Scatterplot with splines showing spending scores vs loyalty points grouped by loyalty category. High loyalty only happens when spending scores are above about 60.

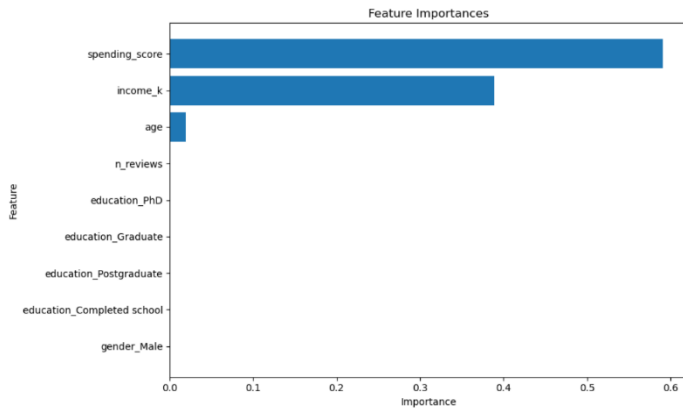


Fig 17: For the decision tree regression model, the feature importances before removing unimportant features.

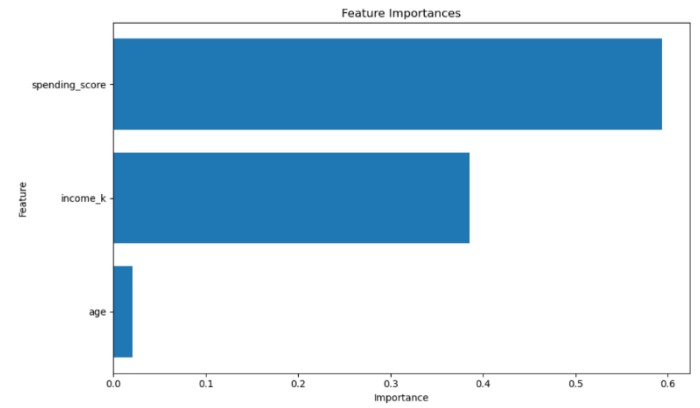


Fig 18: For the decision tree regression model, the feature importances after removing unimportant features.

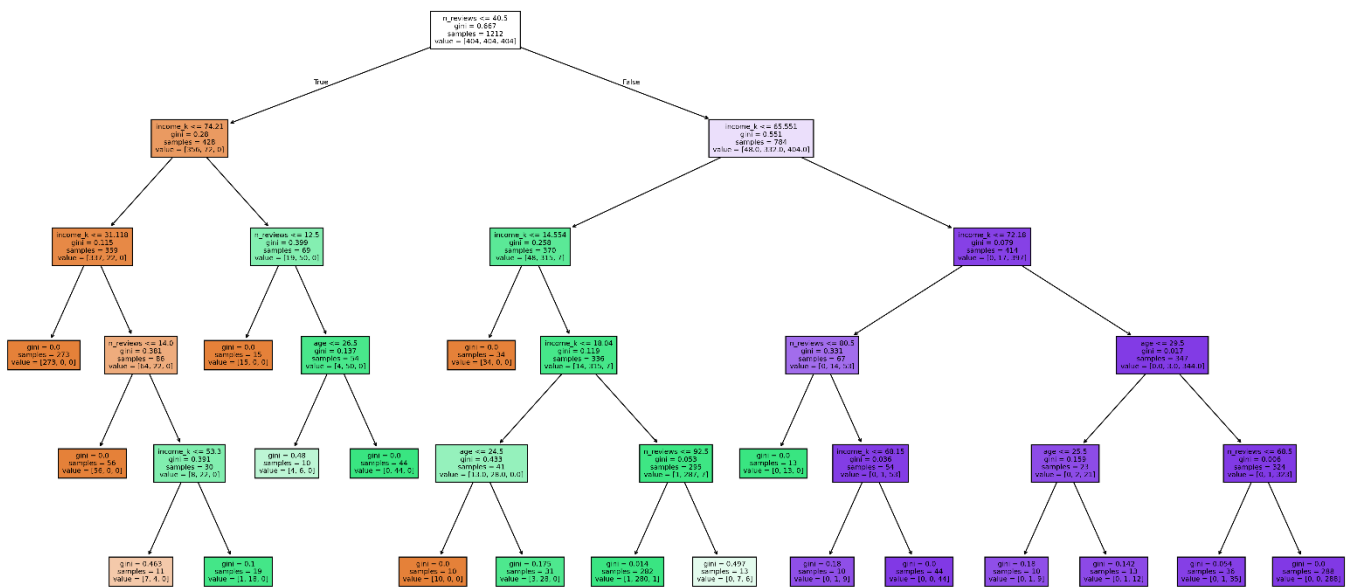
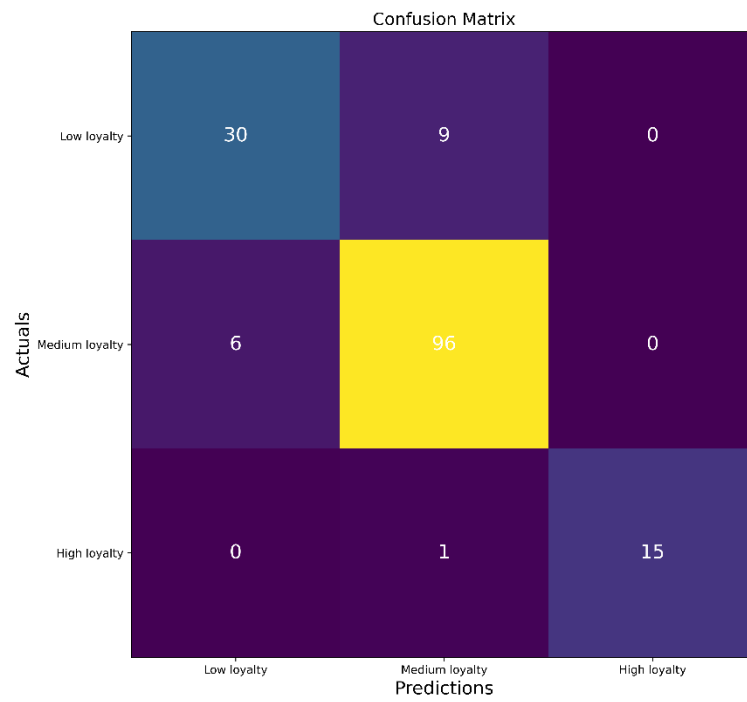


Fig 19: The classification decision tree where age, spending scores, and income are predictors of loyalty points. Purple = High loyalty, Green = Medium loyalty, Orange = Low loyalty. This shows that high spending scores and incomes tend to lead to high loyalty whilst lower spending scores and incomes tend to lead more towards low loyalty and also medium loyalty.



*Fig 20: Confusion matrix for multinomial logistic regression showing that the accuracy of predictions is high.*